Case Study: Accelerating Discovery with Personal AI

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The Problem

National laboratories like SLAC National Accelerator Laboratory sit at the frontier of human knowledge. The science is extraordinary. The operational reality behind it is not.

The Administrative Tax on Discovery

Across U.S. research institutions, scientists spend **42–45% of their time on administrative tasks** rather than doing science. This has remained stubbornly consistent for nearly two decades.

- **22% of research time** consumed by pre-award administration (grant proposals, compliance documentation, budgeting)
- **20% of research time** consumed by post-award administration (progress reporting, financial management, personnel coordination)
- **Grant proposals alone** take an average of **34 working days** per submission
- Administrative overhead spans financial management, IRB/IACUC compliance, time-and-effort reporting, progress reporting, personnel management, and procurement coordination

For a principal investigator running multiple funded projects, this means nearly half of every week is spent on forms, reports, emails, meetings, and documentation.

The Data Deluge

SLAC's scientific instruments generate data at scales that are difficult to comprehend. The existing catalog already spans **60+ petabytes** of research data that scientists must navigate.

The human-scale work that surrounds the science — finding relevant results across years of publications, coordinating distributed teams, managing experimental documentation, writing up findings, communicating results — remains manual.

Why Cloud AI Introduces Unacceptable Risk

Cloud-based AI assistants (ChatGPT, Copilot, Gemini) could help with administrative synthesis, document search, and drafting — but national laboratory environments carry constraints that make cloud AI problematic:

- Export-controlled research
- Pre-publication sensitivity
- DOE data management requirements
- Institutional policy on data handling
- Unauthorized AI use is already happening

The result: researchers who desperately need AI assistance to reclaim their time are either blocked from using it entirely, or using it unsanctioned through consumer tools that offer zero data protection.

Our Solution

FileBuff provides AI-powered research productivity — document search, synthesis, workflow automation, and intelligent communications — with **100% on-device processing**. No research data, proposals, experimental notes, or institutional communications ever leave the device. No cloud AI subscriptions. No third-party data exposure. No new compliance surface.

How It Works for Research Teams

FileBuff integrates with the Microsoft 365 ecosystem (Outlook, Teams, SharePoint, Calendar) used across SLAC and Stanford, delivering capabilities tailored to the research lifecycle:

1. Intelligent Search Across All Research Data

Researchers query across all organizational data sources in plain English. FileBuff's agentic search discovers, previews, extracts, and synthesizes answers — with source citations — across project files, SharePoint document libraries, email archives, and Teams conversations.

Examples at SLAC:

- *"Find all beam time proposals from the last 18 months that used protein crystallography techniques on LCLS"*
- *"What were the key findings from the Q-NEXT quarterly reports on quantum sensing precision?"*
- *"Show me all vendor quotes for cryogenic components received in Q1 and compare pricing"*
- *"Summarize the discussion in the #detector-R&D Teams channel from last week about the new sensor calibration approach"*

What takes hours of manual searching across different systems returns in **minutes** with sourced answers.

2. Automated Research Workflows

FileBuff's visual workflow builder enables lab managers, PIs, and technicians to automate repetitive operational processes — through drag-and-drop or plain language description:

- **Weekly experiment status reports** — Automatically aggregate data from run logs, pull recent results from shared drives, compile into a formatted status report, and

email to the collaboration list every Monday at 0700

- **Equipment maintenance tracking** — Monitor calibration schedules and maintenance logs across instruments, flag overdue items, generate compliance reports for safety review
- **Publication and conference tracking** — Monitor submission deadlines, reviewer feedback, revision due dates, and conference registration windows across the group's active manuscripts
- **Onboarding workflows** — When a new postdoc or student joins, automatically compile required safety training documentation, lab access forms, computing account requests, and orientation materials

3. Document Processing and Synthesis at Scale

Upload hundreds of papers, proposals, reports, or technical documents. FileBuff embeds and indexes them for instant querying.

The AI reads everything, synthesizes across all documents, and returns sourced answers — turning days of literature review into minutes.

4. Intelligent Communications

The AI drafts emails, schedules meetings with availability checking, creates calendar events, and monitors incoming communications:

- **Email triage** — Summarize 100+ daily emails, detect action items, suggest responses in appropriate tone, flag deadlines
- **Meeting preparation** — Pull relevant documents and recent activity before scheduled meetings, generate agenda drafts
- **Collaboration coordination** — Draft status updates from project data, schedule cross-timezone meetings with distributed collaborators, send automated reminders for deliverables

A group leader reviewing their inbox gets AI-generated summaries and suggested responses, spending minutes instead of hours on email — freeing an hour or more each day for actual scientific work.

5. Complete Audit Trail

Every action — human and AI — is logged with full attribution. This supports:

- **DOE compliance** — Auditable record of data access, processing, and sharing
- **Research reproducibility** — Track exactly what queries were run, what documents were processed, and what outputs were generated
- **Institutional review** — Transparent AI usage logging for governance and oversight
- **Export control documentation** — Demonstrate that controlled data was never transmitted externally

Outcomes and Measurable Impact

Operational Performance

Metric	Result
End-to-end workflow (70+ files)	**2.2 minutes**
Semantic search across 500+ documents	**minutes**
AI-drafted email with project context	**10–17 seconds**
Meeting scheduling with availability check	**14 seconds**

Projected Impact Across a Research Division

Metric	Per Researcher	50-Person Division	200-Person Directorate
Admin hours recovered per week	10–15 hrs	500–750 hrs	2,000–3,000 hrs
Admin hours recovered per year	500–750 hrs	25,000–37,500 hrs	100,000–150,000 hrs
Equivalent FTEs returned to research	~0.3 FTE	~15 FTEs	~60 FTEs

At a fully loaded cost of ~$200,000 per researcher-year (salary, benefits, overhead), recovering 15 FTE-equivalents of research time across a 50-person division represents **$3 million in annual productivity recovered** — without hiring a single additional person.

Relevance to SLAC's Mission and DOE Priorities

Accelerating the Researcher

The human bottleneck remains: researchers, PIs, lab managers, and technicians must interpret results, write proposals, coordinate teams, manage compliance, and communicate findings at manual speed.

DOE Data Management and Sharing Plan (DMSP) Compliance

As of October 2025, all DOE-funded research requires approved Data Management and Sharing Plans. FileBuff's architecture supports DMSP requirements by design:

- **Data provenance** — The audit trail logs every AI interaction with research data, creating an auditable chain of custody
- **Access control** — On-device processing means data access is governed by the host machine's existing security controls
- **Reproducibility** — All AI queries, results, and workflow executions are logged for review and replication

DOE Genesis Mission Alignment

The DOE's Genesis Mission aims to create a secure, integrated scientific discovery platform connecting national laboratories, supercomputers, AI systems, and scientific instruments. FileBuff's open-source Obrew platform — with its REST API, modular

architecture, and on-device inference — aligns with this vision as a composable AI layer that any lab or research group can deploy and extend without vendor lock-in or cloud dependency.

Supporting Every Role in the Lab

Role	How FileBuff Helps
Principal Investigator	Automated literature synthesis, grant proposal drafting assistance, weekly project status reports generated from real data, email triage and response drafting
Postdoctoral Researcher	Fast search across institutional knowledge base, experiment documentation assistance, cross-referencing prior results, collaboration coordination
Lab Technician	Equipment workflow automation, calibration tracking, instrument monitoring, automatic documentation
Research Manager	Division-wide status reporting, resource allocation tracking, deadline monitoring, automated meeting preparation
Administrative Staff	Purchase order tracking, travel coordination, onboarding workflows, compliance documentation assembly
Graduate Student	Literature review acceleration, finding relevant prior work, understanding institutional processes, drafting presentations

The Bottom Line

Scientists at SLAC and national laboratories worldwide lose nearly half their working time to administration — searching for documents, writing reports, coordinating teams, managing compliance, and processing email. Cloud AI could help but introduces unacceptable risk for export-controlled, pre-publication, and DOE-governed research data.

FileBuff returns **10–15 hours per researcher per week** to actual science — with 83–96% time savings on administrative tasks, automated workflows, and instant synthesis across hundreds of documents.

Sources

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